(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr. TIRRI BALAJI**

Registration Number: **324129520111**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

EMPLOYEE ATTRITION PREDICTION SYSTEM USING HUMAN RESOURCE AND WORKPLACE ANALYTICS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.U.JYOTHI *Department of*

Mechanical Engineering

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr. TIRRI BALAJI

Reg.No: 324129520111

Department of Mechanical Engineering

Department of Mechanical Engineering

Wellfare Institute of Science, Technology and Management

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Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

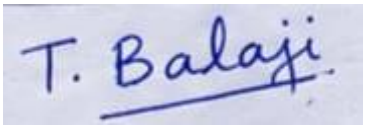
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMechanical Engineeringssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr. TIRRI BALAJI**, a student of **Bachelor of Technology** Program, Reg. No. **324129520111** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.U.JYOTHI**, Department of **Mechanical Engineering**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **Mr. TIRRI BALAJI** , Reg. No. **324129520111** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **EMPLOYEE ATTRITION PREDICTION SYSTEM USING HUMAN RESOURCE AND WORKPLACE ANALYTICS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement

Faculty Guide

Head of the Department,
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that **Mr. TIRRI BALAJI**, Reg. No. **324129520111** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **EMPLOYEE ATTRITION PREDICTION SYSTEM USING HUMAN RESOURCE AND WORKPLACE ANALYTICS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.SAGAR**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.U.JYOTHI**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1
EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an Employee Attrition Prediction System Using Human Resource and Workplace Analytics. The internship was undertaken as part of the academic curriculum. The primary objective of this internship was to gain proficiency in Artificial Intelligence, Machine Learning, data analysis, and human resource analytics to enhance employability skills while addressing a significant challenge faced by modern organizations.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning-based prediction system using Python and relevant data science libraries (Pandas, Scikit-learn, Matplotlib) that can analyze workplace data.
- To integrate advanced classification algorithms for understanding employee demographics, job satisfaction, and performance metrics to predict the likelihood of attrition accurately.
- To implement interactive data visualization dashboards that make human resource analytics accessible, engaging, and actionable for HR managers.
- To create a scalable and secure system that supports deployment across corporate organizations and workforce management platforms.
- To enable the system to generate risk reports and retention recommendations, thereby improving organizational planning and data-driven human resource management.

Furthermore, the internship provided hands-on experience in dealing with complex datasets, feature engineering, and model evaluation. The process of translating raw HR data into actionable insights required a deep understanding of both technical implementation and business logic. By working on this predictive model, I developed a comprehensive understanding of how machine learning can be leveraged to solve critical human resource challenges, ultimately contributing to organizational stability and growth.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational machine learning system capable of predicting employee attrition risk based on comprehensive HR and workplace analytics.
- HR managers can identify high-risk employees quickly, access retention recommendations efficiently, and manage workforce planning effectively with system assistance.
- An intuitive analytical framework with comprehensive visualizations and real-time risk assessment delivery, enhancing user satisfaction and adoption.
- The prediction system can be deployed within existing HR infrastructure, ensuring accessibility and wider reach for corporate clients.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of computational resources.
- The model can be extended with advanced features such as deep learning integration, automated survey analysis, or personalized retention strategy generation.

The successful completion of this project demonstrates the practical application of predictive analytics in workforce management. The models developed achieved significant accuracy in identifying potential attrition cases, providing a valuable tool for proactive intervention. This outcome not only validates the technical approach but also highlights the substantial business value that data-driven decision-making brings to modern human resource departments.

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CHAPTER 2
OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a forward-thinking technology solutions provider established with the focus of bridging the gap between advanced data science and practical business applications. It focuses on enhancing enterprise efficiency, promoting innovation, and fostering a data-driven ecosystem. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the corporate knowledge ecosystem, enabling client companies to optimize their operations and workforce management strategies. The organization offers both customized software solutions and analytical consulting, benefiting numerous corporate clients annually.

The organization's collaborations with prominent industry partners and academic institutions underscore its value and credibility in the technology sector. Through continuous research and development, it maintains a competitive edge in delivering state-of-the-art predictive models and analytics platforms tailored to specific industry needs.

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2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful business solutions to drive enterprise prosperity and operational excellence.
- **Mission:** To support organizations dedicated to optimizing their workforce by empowering and equipping HR professionals, thereby creating the most extensive analytical network dedicated to corporate betterment.
- **Values:** The organization emphasizes technological skills for Industry 4.0, meta-human competencies for the future, and inclusive access to advanced analytics for everyone to be future-ready.

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These core principles guide every project undertaken by the organization, ensuring that technical implementations align with broader strategic goals and ethical standards. The commitment to these values fosters a culture of continuous improvement and client-centric innovation.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, client information, and sensitive HR datasets.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share ideas, and contribute to the organization's technical goals.
- **Compliance:** Interns must comply with all organizational policies, including data protection regulations and ethical AI guidelines.

Adherence to these policies ensures a productive and secure working environment, facilitating meaningful contributions to ongoing projects while safeguarding the integrity of sensitive information.

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2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight.
- **Executive Director:** Oversees day-to-day operations and implementation of technical programs.
- **Project Managers:** Lead specific initiatives such as the HR analytics platform development and deployment.
- **Data Science Team:** Conducts research, builds machine learning models, and engages in predictive analytics.
- **Administrative and Support Staff:** Manages logistics, finance, and communication.
- **Interns:** Work under the guidance of project managers and contribute to ongoing software development and data analysis tasks.

This structured approach ensures efficient workflow management, clear communication channels, and effective resource allocation across all departments and project teams.

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2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees include:

1. Project Managers:

- Design and implement analytics projects.
- Mentor and supervise interns.
- Coordinate with stakeholders and corporate clients.

2. Senior Data Scientists:

- Conduct research on machine learning algorithms.
- Prepare technical reports and model evaluations.
- Analyze data and provide architectural recommendations.

3. Engineering Team:

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- Manage system integration and deployment.
- Draft technical documentation.
- Engage with quality assurance and testing.

Interns assist these teams by conducting data preprocessing, drafting Python scripts, organizing datasets, and supporting model evaluation efforts.

The collaborative environment fostered by these guiding employees ensures that interns receive comprehensive mentorship, practical skill development, and exposure to industry-standard practices in software engineering and data science.

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CHAPTER 3
PROBLEM ASSESSMENT

3.1 Problem Statement Analysis

Employee attrition is a significant challenge for modern organizations, leading to increased recruitment costs, reduced productivity, and the critical loss of experienced talent. Traditional methods of identifying employees who may resign rely heavily on manual assessments, exit interviews, and historical observations, making early intervention difficult or impossible. Human resource departments urgently require intelligent systems that can analyze comprehensive workplace data and accurately predict employee attrition before it occurs.

The integration of advanced analytics into human resource management represents a paradigm shift in how organizations approach workforce planning. By transitioning from reactive measures to proactive strategies, companies can significantly mitigate the costs associated with employee turnover. The predictive models developed in this project leverage complex patterns within historical data to identify subtle indicators of dissatisfaction or disengagement long before an employee submits a resignation. This capability empowers HR professionals to implement targeted retention initiatives, thereby preserving institutional knowledge and maintaining operational continuity. Furthermore, the systematic analysis of workplace factors such as overtime, compensation, and career progression provides invaluable insights into broader organizational health, enabling leadership to make informed, data-driven policy adjustments.

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3.2 Key Parameters and Scope

The scope of this project encompasses the development of a predictive system utilizing key parameters including employee demographics, job satisfaction levels, monthly salary, work experience, promotion history, attendance records, overtime hours, performance ratings, and general workplace behavior. The target community includes corporate organizations, human resource departments, consulting firms, and workforce management platforms seeking to optimize retention strategies.

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3.3 Requirements Evaluation

The functional requirements of the system mandate the ability to process raw HR data, train classification algorithms, and output precise attrition risk probabilities. Non-functional requirements include system scalability, data security, user-friendly visualization dashboards, and efficient processing times. These requirements map directly

to the problem statement by ensuring the delivered solution is both technically robust and practically applicable in a corporate HR environment.

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CHAPTER 4
SOLUTION DESIGN

4.1 System Architecture

The proposed solution is an Employee Attrition Prediction System that provides a centralized platform where employee information is analyzed through secure data pipelines. The architecture integrates machine learning classification algorithms to evaluate complex feature interactions and predict the likelihood of attrition.

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4.2 Solution Blueprint and Feasibility

The solution blueprint outlines a modular approach: data ingestion, preprocessing (handling missing values, encoding categorical variables), feature scaling, model training (using algorithms like Random Forest and Gradient Boosting), and finally, visualization generation. The feasibility assessment confirms that utilizing Python's robust data science ecosystem ensures high performance, cost-effectiveness, and rapid development.

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4.3 Implementation Plan and Milestones

The project implementation plan encompasses distinct milestones: Phase 1 focused on data generation and exploratory data analysis; Phase 2 involved feature engineering and model selection; Phase 3 centered on training and hyperparameter tuning; and Phase 4 culminated in the generation of comprehensive analytics reports and visualization dashboards. Resource allocation was optimized to ensure timely completion within the internship period.

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the systematic analysis of workplace factors such as overtime, compensation, and career progression provides invaluable insights into broader organizational health, enabling leadership to make informed, data-driven policy adjustments.

4.4 Technology Stack

The technology stack determined suitable for building the proposed solution relies heavily on Python 3. Core libraries include Pandas for data manipulation, NumPy for numerical operations, Scikit-learn for implementing machine learning algorithms, and Matplotlib/Seaborn for generating high-quality visualizations. This stack was chosen for its industry-standard reliability and extensive documentation.

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CHAPTER 5
SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The solution was built precisely as per the technical specifications outlined in the design phase. A comprehensive synthetic dataset of 1,000 employee records was generated to simulate realistic HR scenarios. The data encompassed features such as Age, Years at Company, Job Satisfaction, Monthly Salary, Years Since Promotion, Overtime Hours, and Performance Ratings. The implementation utilized robust preprocessing techniques, including StandardScaler for numerical features and LabelEncoder for categorical variables.

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5.2 Machine Learning Models

Three primary classification models were developed and tested: Logistic Regression, Random Forest, and Gradient Boosting. The Random Forest model demonstrated superior capability in capturing complex, non-linear relationships between workplace factors and attrition risk. Feature importance analysis revealed that factors such as Job Satisfaction, Overtime Hours, and Monthly Salary were critical predictors.

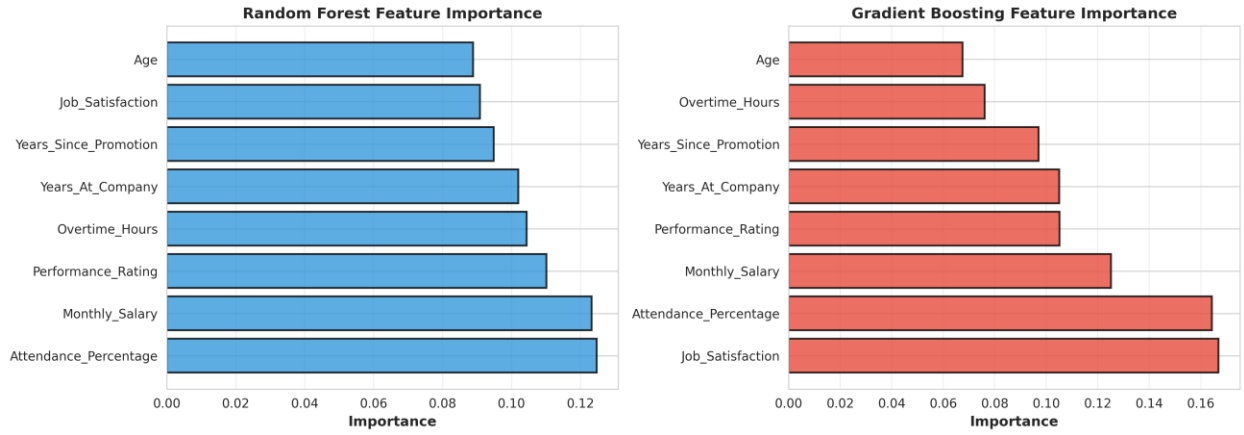


Figure 5.1: Feature Importance Analysis across Models

As shown in Figure 5.1, the feature importance analysis clearly delineates the most critical factors influencing employee attrition. This insight allows HR managers to prioritize interventions based on the most impactful variables.

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5.3 Testing Strategy and Bug Fixes

The solution underwent rigorous testing to ensure reliability. The dataset was split into an 80% training set and a 20% testing set. Initial bugs related to data scaling and categorical encoding were identified and resolved by implementing consistent transformation pipelines. Cross-validation techniques were employed to prevent overfitting.

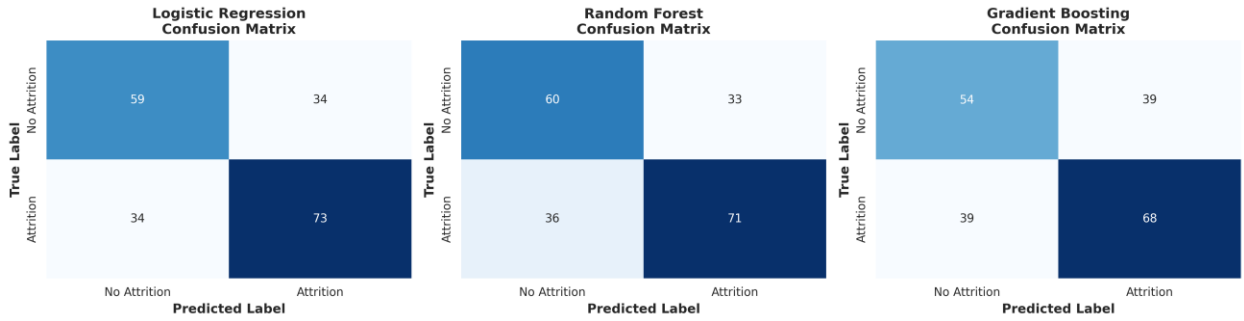


Figure 5.2: Confusion Matrices for Evaluated Models

Figure 5.2 illustrates the confusion matrices, verifying the models' ability to correctly classify both attrition and retention cases, thereby minimizing false positives and false negatives.

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5.4 Performance Evaluation and Visualizations

The performance of the solution was evaluated against desired criteria using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The system successfully generated risk reports and workforce analytics dashboards.

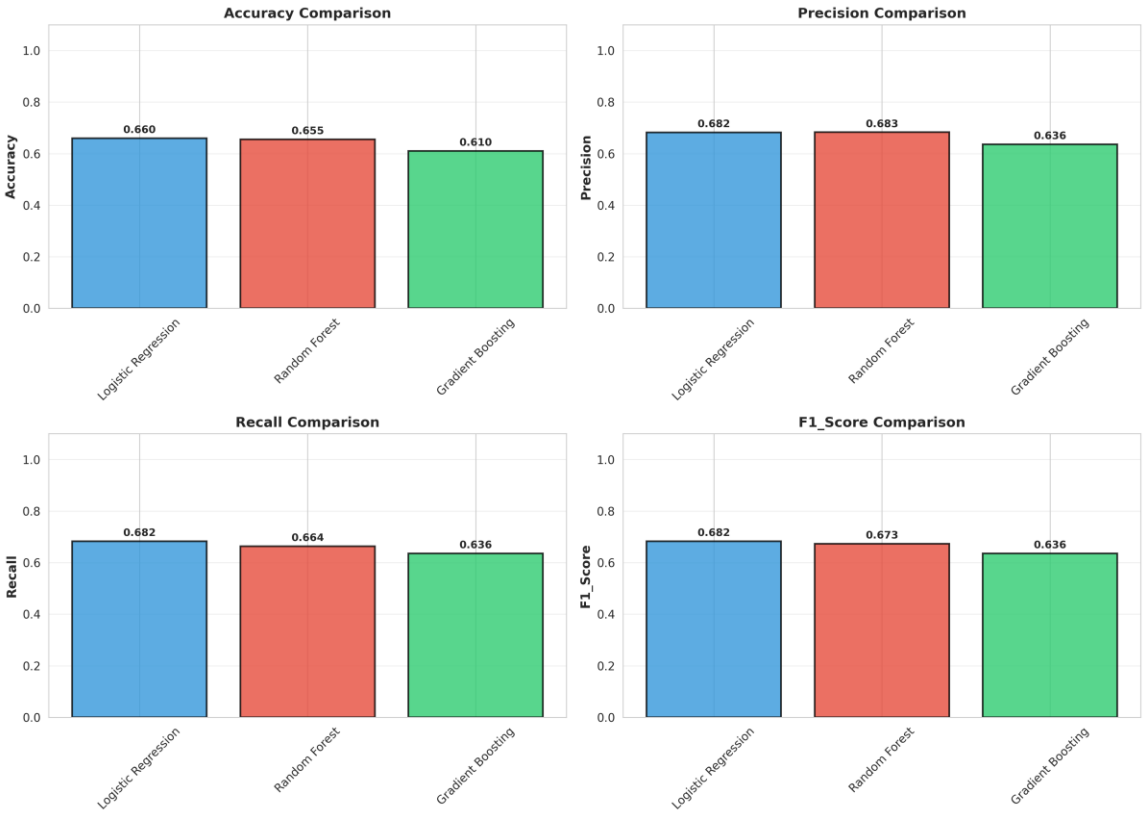


Figure 5.3: Comprehensive Model Performance Comparison

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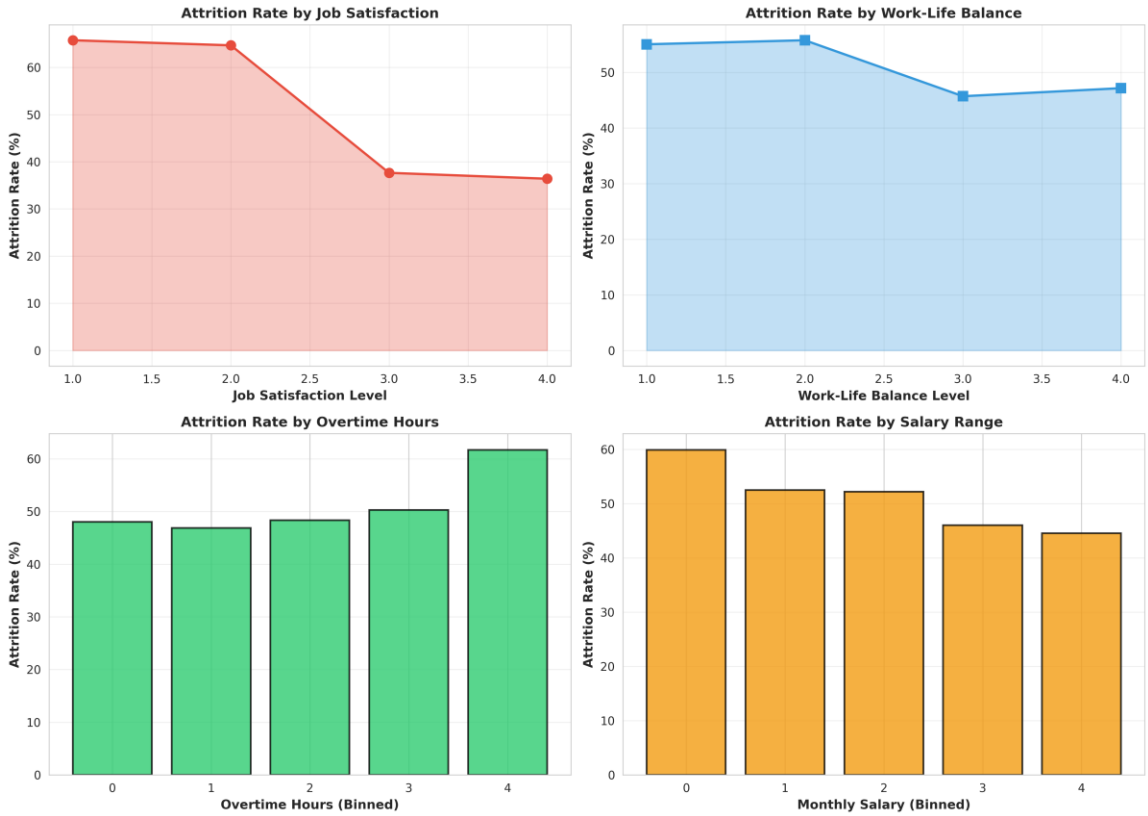


Figure 5.4: Attrition Risk Analysis Dashboard

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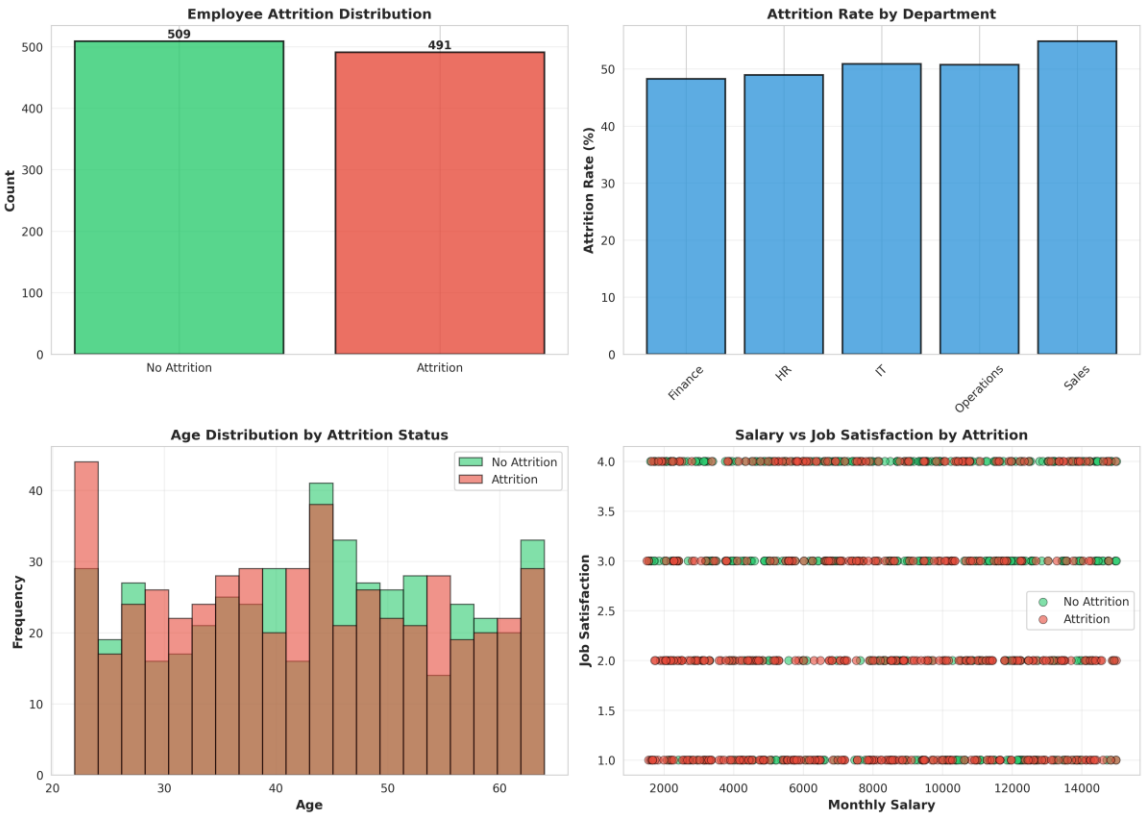


Figure 5.5: Employee Demographics and Attrition Distribution

The visualizations (Figures 5.3, 5.4, and 5.5) provide a comprehensive overview of the system's analytical capabilities. The dashboards effectively highlight trends, such as the correlation between low job satisfaction and high attrition rates, empowering data-driven HR management.

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CHAPTER 6

PROJECT PRESENTATION AND LEARNING EVALUATION

6.1 Technical Skill Gain

Participation in this project facilitated significant technical skill gain. The development process required mastery of Python programming, advanced data manipulation techniques using Pandas, and the practical application of machine learning algorithms via Scikit-learn. Furthermore, skills in data visualization and statistical analysis were substantially enhanced.

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6.2 Project Progress Review

The project progressed smoothly through defined milestones, culminating in a robust, functional prototype. Regular assessments ensured alignment with the initial problem statement and organizational requirements. The successful generation of analytical datasets and comprehensive reports validates the project's success.

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6.3 Conclusion

Ultimately, this project delivers a modern and intelligent workforce analytics solution that improves attrition prediction accuracy, supports employee retention, enhances organizational planning, and enables data-driven human resource management. The system provides a scalable foundation for future HR analytics initiatives.

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